

Assignment Lab 5.2 — README

Instructions

Course 02 – Introduction to Machine Learning for Cybersecurity

Module 05: Introduction to Neural Networks

Assignment Lab 5.2

Modify and Evaluate a Neural Network Model

Lab Goal

In this assignment, you will build on Guided Lab 5.1 by modifying and evaluating a neural network model using TensorFlow and Keras.

You will use the `creditcard.csv` dataset to classify transactions as:

- `Class = 0` → Normal transaction
- `Class = 1` → Fraudulent transaction

This assignment focuses on understanding how changes to a neural network model can affect performance.

Note

The goal of this assignment is not only to build neural network models, but also to understand:

- How neural networks learn
- How architecture changes affect predictions
- How model performance is evaluated

- How AI systems support cybersecurity operations



Tips for Success

- Carefully compare both neural network models
- Observe how architecture changes affect performance
- Pay attention to precision and recall
- Think critically about false positives and false negatives
- Connect your findings to cybersecurity and fraud detection contexts



Cybersecurity Connection

Neural networks are increasingly used in:

- Fraud detection systems
- Intrusion detection systems
- Malware classification
- Threat intelligence systems
- User behavior analytics

This assignment demonstrates how neural networks can support cybersecurity and fraud detection workflows by identifying suspicious activity patterns.



Dataset

Use the provided dataset:

✓ `creditcard.csv`

[creditcard.csv](#)

Remember to update the dataset path in your notebook if the file is saved in a different location.



Important Notes

- Ensure TensorFlow is installed before running the notebook
- Update the dataset path if needed
- Use comments to explain your code and workflow
- Focus on interpreting results, not only generating outputs

- Remember that fraud detection datasets are highly imbalanced, so accuracy alone may be misleading

✂ What You Will Do

You will:

1. Load and explore the dataset
2. Prepare the data for neural network modeling
3. Build a baseline neural network model
4. Modify the neural network architecture
5. Train and evaluate both models
6. Compare model performance
7. Adjust the prediction threshold
8. Interpret the results in a fraud detection and cybersecurity context

Required Tasks

Part 1 — Import Libraries

Import the required Python, Scikit-learn, TensorFlow, and Keras libraries.

Part 2 — Load and Explore the Dataset

Load `creditcard.csv`, review the dataset shape, check for missing values, and examine the class distribution.

Part 3 — Prepare the Data

Students should:

- Define features and target variables
- Split the dataset into training and testing sets
- Scale the features using `StandardScaler`

Part 4 — Build a Baseline Neural Network

Create a simple neural network model using TensorFlow/Keras.

Example Components:

- Dense layers
- ReLU activation
- Sigmoid output layer

Part 5 — Train and Evaluate the Baseline Model

Train the model and evaluate it using:

- Accuracy
- Precision
- Recall
- F1 Score
- Classification Report
- Confusion Matrix

Part 6 — Build a Modified Neural Network

Modify the neural network by changing one or more of the following:

- Number of hidden layers
- Number of neurons
- Dropout layer
- Batch size
- Number of epochs
- Prediction threshold

Part 7 — Compare Model Performance

Compare the baseline and modified neural network using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion matrices
- Training history visualizations

Part 8 — Reflection

Answer the reflection questions included in the notebook.



Reflection Questions

Answer the following questions in complete sentences:

1. Which neural network model performed better overall?
2. Did the modified model improve precision, recall, or F1 score?
3. Why is accuracy alone not enough for fraud detection?
4. What did the confusion matrices show?
5. How did changing the prediction threshold affect the results?
6. What are the strengths of neural networks for cybersecurity applications?
7. What are the limitations of neural networks in fraud or cybersecurity detection?
8. How could this model be improved in future work?



Submission Requirements

Submit the following in Canvas:

- ✓ Completed Jupyter Notebook (.ipynb)
- ✓ All code cells executed with outputs visible
- ✓ Evaluation metrics included
- ✓ Confusion matrices included
- ✓ Model comparison table included
- ✓ Training history plots included
- ✓ Reflection questions answered